Do Now · retrieval

Lesson 082 · Quicksort: Partition Around a Pivot

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is merge sort's combine step?

Q2

What is a pivot in quicksort?

Q3

Is a partitioned array necessarily fully sorted?

Q4

Swap values at indexes0 and2 in [4,1,7]. Give the resulting array.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Observe — Partitioning reorders values around a chosen pivot so values to its left are no larger and those right are larger, for this partition rule.

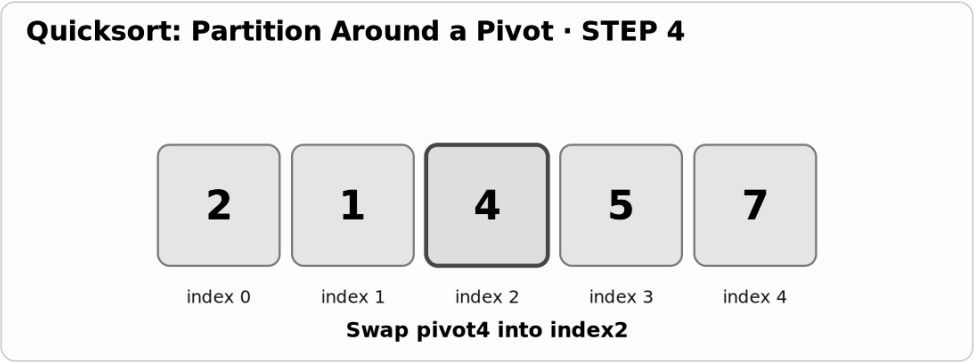
Core idea 2

Apply — The pivot's final index is fixed after partition, but neither side is necessarily fully sorted; quicksort continues with the two subarrays.

Core idea 3

Evaluate — Average and best-case time O(n log n), worst $O(n^2)$ with poor pivot choices; recursion consumes stack space depending on partition balance.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

quicksort — Recursive partition-based sorting.

pivot — Value partitioning subarray.

partition — Place smaller/larger values around pivot.

worst case — Maximum growth for adverse input.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

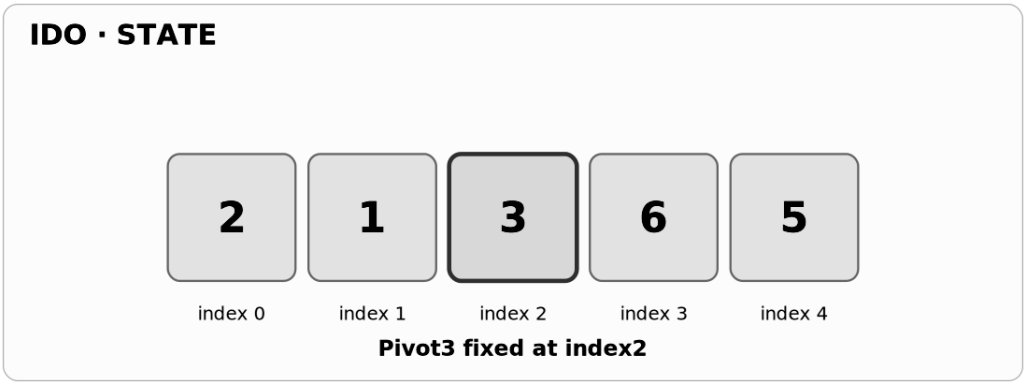
Problem

Partition [6,2,5,1,3] using last pivot3; give partition state.

Follow the worked method

1. Method and result — Lomuto partition moves 2 and 1 left, pivot3 to index2 → [2,1,3,6,5].

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Method and result
Smaller 3,2 move left; pivot5 at index2 → [3,2,5,8,6].

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Trace Lomuto pivot-last partition of [5,2,7,1,4] and give final pivot index. [6]

Question 2

After quicksort partition gives [2,1,4,5,7] with pivot4 at index2, state required recursive subranges and eventual sorted result. [6]

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain how pivot choice impacts quicksort worst-case time and recursive stack depth. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Once a pivot reaches its correct final index, both partitions are also completely sorted.

Use key vocabulary from memory

1. Explain quicksort in your own words.

2. Explain pivot in your own words.

3. Explain partition in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.